

Jharkhand University of Technology, Ranchi

Diploma 5th Semester Examination, 2024

Subject : Computer Graphics

Subject Code : CSE-505

Time Allowed : 3 Hours

Full Marks : 80

Pass Marks : 26

Answer in your own words.

Answer five questions in which Question No. 1 is compulsory.

The figures in the margin indicate full marks.

All question carry equal marks.



Choose the correct option:

2×8=16

(i) Which of the following is a line drawing algorithm?

- (a) Line equation algorithm
- ☒ (b) Digital Differential Analyzer (DDA) algorithm
- (c) Bresenham's algorithm
- ☒ (d) All of the above

(ii) Expansion of line DDA algorithm is

- ☒ (a) Digital Difference Analyzer
- ☒ (c) Digital Differential Analyzer
- (b) Direct Differential Analyzer
- (d) Data Differential Analyzer

(iii) An efficient transformation method which produces a parallel mirror image of an object is also referred as,

- (a) Rotation
- ☒ (b) Reflection
- (c) Shear
- (d) All of these

(iv) RGB model are used for _____.

- ☒ (a) computer display
- (b) printing
- (c) painting
- (d) None of these

(v) _____ Transformation alters the size of an object.

- ☒ (a) Scaling
- (b) Rotation
- (c) Translation
- (d) Shear

(vi) What is the formula for calculating the slope 'm' of a line?

- (a) $m = \frac{dx}{dy}$
- ☒ (b) $m = \frac{dy}{dx}$
- (c) $y = mx + c$
- (d) None of these

$$y = mx + b$$

(vii) Which one of the following is the correct notation of a matrix with 'm' rows and 'n' columns?

(a) $m + n$

(b) $m - n$

✓(c) $m \times n$

(d) $\frac{m}{n}$

(viii) On raster system, lines are plotted with

(a) lines

(b) Dots

✓(c) pixels

(d) None of these

2. (a) Describe CRT display. Explain its working with diagram.

(b) Magnify the triangle with vertices A (0, 0), B (1, 1), and C (5, 2) to twice its size while keeping C (5, 2) fixed. 8+8

3. (a) Explain the steps of Bresenham's Line drawing Algorithm. Using this algorithm, draw a line with the end point: (10, 10) and (60, 40).

(b) Draw a circle with radius 20 and centre is at (0, 5) by applying mid-point circle drawing algorithm. 8+8

4. (a) Find the transformation matrix that transforms the square ABCD whose center is at (2, 2) is reduced to half of its size, with center still remaining at (2, 2). The co-ordinate of square ABCD are A (0, 0), B (0, 4), C (4, 4) and D (4, 0). Find the co-ordinate of new square.

(b) Apply Clipping, a line (-1, 5) and (3, 8) using the Cohen Sutherland algorithm with Window coordinates (-3, 1) and (2, 6). 8+8

5. (a) Differentiate between Random Scan and Raster Scan Displays.

(b) Explain Back face detection with suitable example. 8+8

6. (a) How to represent Curve and Fractals. Explain any two techniques.

(b) Explain the color models define with primary colors. 8+8

7. Write short notes on any four:

(a) B-Spline Curves

✓(b) Boundary Fill Algorithm

✓(c) Raster Scan Display

✓(d) Shearing Transformation

✓(e) Painter's algorithm

4×4=16